

Peter Howe

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Experience

UBISOFT

Montreal, Canada

Online Services SDK Programmer, Term 3

Jan 2026 – Apr 2026

- Overhauled **WinHTTP** implementation by decoupling the **WebSocket** and **HTTP** implementations and replacing global mutexes with fine-grained locks, improving concurrency and reducing thread contention
- Modernized a legacy **CI/CD** system by consolidating 140+ scripts into a single **CMake** Presets configuration, simplifying build orchestration and maintainability
- Refactored a 6,000-line SWIG interface into 75 modular components, **reducing code size by 96%** and enabling scalable, maintainable **C#** API integration
- Replaced legacy **HTTP** engine string implementation with a copy-on-write design using move semantics, improving performance and memory efficiency; validated via **Clang-Tidy** CI checks
- Collaborated with platform SDKs (PlayStation, Xbox, Android) to implement production-ready cross-platform features

Online Services SDK Programmer, Term 1 & 2

Sept 2024 – Apr 2025

- Redesigned SWIG-generated **C#** bindings across 11 platforms, transitioning from standalone clients to a modular architecture while preserving backward compatibility with **Google Test**
- Diagnosed and resolved critical crashes affecting *Rainbow Six Siege Mobile* on Android
- Improved platform flexibility by introducing configurable web renderer selection across console platforms without breaking existing APIs

CONCORDIA STUDENT SUCCESS CENTER

Montreal, Canada

Lead Developer

Aug 2024 – Present

- Conceived and developed a specialized GenAI tutoring system that incorporates course knowledge and context to curate a personalized tutoring experience, made available to **thousands of students** annually
- Independently developed a full-stack application in **Python/Flask** and **Typescript/React**, handling student data with **PostgreSQL**, authentication with **Azure AD**, and deployment with **Docker**
- Presented the platform to the Learning Specialist 2025 Conference and internally at Concordia

Peer Tutor

Sept 2023 – Apr 2025

- Provided 1-on-1 tutoring in calculus and linear algebra, helping students build foundational math skills for upper-level science and engineering courses

Education

CONCORDIA UNIVERSITY

Computer Science, GPA 3.8/4.3

Montreal, Canada

Graduating December 2026

Extracurricular Activities

SPACE CONCORDIA

Sept 2023 – Sept 2024

- Embedded development in ADA/SPARK for SC-FREYR, a Canadian Space Agency-funded CubeSat

CHESS ENGINE PROJECT

- Built a C++ Chess Engine implementing Minimax search, Alpha-Beta Pruning, and Quiescent Search

Skills

Programming Languages: C++, C#, Golang, Java, Python, Typescript/Javascript, Lua, Bash, Batch

Frameworks: Flask, React, PostgreSQL, Azure AD

Tools: CMake, GitLab CI, Git, Google Test, Swig, Visual Studio, Linux, Vim

Language: English, French